

Nicolas DAVAL

Data Architect & Tech Lead

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PROFESSIONAL SUMMARY

Data Architect & Tech Lead with 22+ years of banking experience (Stellantis Financial Services), specialized in designing and industrializing enterprise-grade cloud data platforms at multi-country scale. Co-leads the Snowflake / AWS / dbt transformation program serving 8 European countries, with standards (RBAC, governance, privacy, data contracts) adopted as the SFS Banking reference. Designs ingestion and transformation frameworks that cut data flow time-to-market (x4 target) and accelerates delivery through intensive use of GenAI and Kiro. Long-standing software architect — introduced microservices and event-driven patterns at SFS, designed an Ops industrialization framework still in production today. Known for owning architectural trade-offs, managing crises (UK regulatory, platform performance) and aligning business, IT and partners across 8 countries.

TECHNICAL SKILLS

Data & Cloud: Snowflake, dbt, Modélisation 3NF, Data Vault V2, Airflow, Astronomer Cosmos, DAG Factory, Elementary, AWS (S3, Glue, DMS, API Gateway), Python, SQL

Development & Architecture: Java / Spring / Spring Cloud, Architecture Event-Driven / CQRS, Kafka / RabbitMQ, MongoDB, Oracle / PostgreSQL / MySQL, Docker, Shell / Perl, API REST

Data Quality & Governance: Elementary (Data Quality), Data Governance, Privacy / RGPD

GenAI & Innovation: GenAI / MCP, Kiro (Steerings, Skills, Hooks, Requirements), YAML

Tools & Methodologies: Git / Version Control, CI/CD, TeamCity, Github Actions, Agile / Scrum

PROFESSIONAL EXPERIENCE

Data Tech Lead / Data Architect

2021 - Present

Stellantis Financial Services

- Co-led a multi-country data platform transformation program (8 European countries) on Snowflake / AWS / dbt — CIO + Chief Data Officer sponsorship, ~20-person platform team
- Defined target architecture with architects — Snowflake RBAC, MWAA, AWS Glue EL platform, 3NF then Gold-layer Data Products, data contracts introduction
- Designed and industrialized an EL framework and a bank-grade generic extractor (Python + dlthub) — Bronze flow onboarding from full manual build to 1.5 days, x4 delivery target, adopted on 100+ sources per country
- Defined SFS Banking standards (privacy, governance, conventions, quality) co-built with the architect and Data Platform Manager
- Owned and defended architectural trade-offs (e.g. enforced removal of a cross-account Glue DB pull that breached enterprise integration patterns)

Technologies: Snowflake, AWS (S3, Glue, MWAA), dbt, Airflow, Astronomer Cosmos, DAG Factory, Elementary, Python, dlthub, Modélisation 3NF / Data Vault, Data contracts, GenAI / Kiro, SQL

Software Architect & Innovation

2016 - 2021

Stellantis Financial Services

- Architected the digital banking platform exposing core banking to customer, sales and back-office channels across 8 countries — 30+ consumers, up to 12M requests per month at peak
- Introduced and evangelized microservices and event-driven patterns at SFS (previously non-existent) — application boundaries aligned on the BIAN business framework, ~10 Spring Boot microservices
- Led the refactoring and cloud migration of the on-prem platform (Docker / Ansible) to AWS — continuous FinOps approach (night mode, automatic pre-prod shutdown, runtime optimization)
- Contributed to the Enterprise Business Model definition (BIAN, Open Banking, SFS architecture reference)
- Innovation — first SFS AI PoC on IBM Watson, chatbot solutions evaluation

Technologies: Java / Spring Boot / Spring Cloud, MongoDB, Kafka, RabbitMQ, Docker / Ansible, AWS, BIAN, IBM Watson, Hadoop / Spark

Java Tech Lead

2010 - 2016

Stellantis Financial Services

- Software architect and Tech Lead on the end-to-end 4x interest-free payment solution (front office + back office, collections, accounting) — rolled out to ~7,500 dealers, thousands of requests per month, still live today
- First SFS Java application with front / back separation despite using Struts 2 — pattern became an internal reference for later Java initiatives
- Supervised and coordinated an offshore development team (China)
- Continuous framing and stabilization of a complex business need on collections and accounting
- Designed and delivered a home-made Java ETL serving as a batch Mediation Layer

Technologies: Java / JEE, Spring, Struts 2, Oracle, PostgreSQL, Shell, SQL

Application Integrator

2002 - 2010

Stellantis Financial Services

- Designed and delivered a Shell / Perl industrialization framework rolled out across the full SFS scope — application onboarding and go-live reduced from several months to 1 week, framework still in production today
- Defined and rolled out the SFS Version Control strategy — first Git adoption on the Integration side, packaging tooling, change approval committee setup
- Integrated Java applications on Unix / Linux and Windows across the full banking scope
- Actively contributed to the evolution of the bank's technical architecture

Technologies: Shell, Perl, Java, Unix / Linux, Windows Server, Git

Network Teacher (BTS level) ICEP	2003 - 2003
• Teaching computer networks at BTS level (2-year technical degree)	
Apprentice — GIS Laboratory CNERTA Dijon	2001 - 2002
• Geographic Information Systems laboratory using ESRI software	
• Development of an embedded tool for Windows Pocket PC	
<i>Technologies: PostgreSQL, Perl, ESRI (ArcGIS), Windows Mobile</i>	
Intern / Apprentice — Intranet Établissements Schiever	2000 - 2001
• Design and deployment of an intranet for stores to place orders and browse the product catalog	
<i>Technologies: Web development</i>	

EDUCATION

Engineering Degree (Master's equivalent) UTBM	2006
Master's Degree in Computer Science CNAM	2004
CASIE — Enterprise Information Systems Designer & Administrator ICEP Dole	2002
DUT in Computer Science (2-year technical degree) IUT Dijon	2000
Scientific Baccalaureate Lycée d'Avallon(89)	1998

PROFESSIONAL TRAINING

- Snowflake Starter Kit (Snowflake)
- MongoDB University — Parcours Développeur Java (MongoDB)
- AWS Solutions Architect (AWS)
- AWS DevOps (AWS)
- AWS Go Build Data & IA (AWS)
- Spring Framework (Orsys)
- Angular (Orsys)
- Oracle 9 (Oracle)

SOFT SKILLS

- Banking & data expertise: 22+ years in banking (Stellantis FS) — BIAN framework, regulatory environment, Central Bank and ISO audits
- International team management: Teams distributed across multiple sites (France, Argentina, China, India) — inshore, nearshore, offshore
- Communication & stakeholder buy-in: Ability to simplify complex topics, challenge ideas and drive stakeholder buy-in
- Crisis management: Driving sensitive and urgent regulatory topics in taskforce / war room mode
- Data evangelism: Promoting data culture and best practices within the organization

LANGUAGES

- Français — Langue maternelle
- Anglais — Professionnel